```
import pandas as pd

data = {
    "Customer_ID": [101, 102, 103, 104, 105, 106, 107, 108],
    "Name": ["Rahul Mehta", "Anjali Rao", "Suresh Iyer", "Neha Singh", "Amit Verma", "Karan Shah", "Pooja Das", "Riya Kapoor
    "City": ["Mumbai", "Bengaluru", "Chennai", "Delhi", "Pune", "Ahmedabad", "Kolkata", "Jaipur"],
    "Monthly_Sales": [12000, None, 15000, None, 18000, None, 14000, 16000],
    "Income": [65000, None, 72000, None, 58000, 61000, None, 69000],
    "Region": ["West", "South", "South", "North", None, "West", "East", "North"]
}

df = pd.DataFrame(data)
df.to_csv("customer_dataset.csv", index=False)
```

```
df
```

	Customer_ID	Name	City	Monthly_Sales	Income	Region
0	101	Rahul Mehta	Mumbai	12000.0	65000.0	West
1	102	Anjali Rao	Bengaluru	NaN	NaN	South
2	103	Suresh Iyer	Chennai	15000.0	72000.0	South
3	104	Neha Singh	Delhi	NaN	NaN	North
4	105	Amit Verma	Pune	18000.0	58000.0	None
5	106	Karan Shah	Ahmedabad	NaN	61000.0	West
6	107	Pooja Das	Kolkata	14000.0	NaN	East
7	108	Riya Kapoor	Jaipur	16000.0	69000.0	North

Q8. Listwise Deletion Remove all rows where Region is missing.

Tasks:

```
#1. Identify affected rows
region_drop = df.dropna(subset=['Region'])

region_drop
```

	Customer_ID	Name	City	Monthly_Sales	Income	Region
0	101	Rahul Mehta	Mumbai	12000.0	65000.0	West
1	102	Anjali Rao	Bengaluru	12000.0	NaN	South
2	103	Suresh Iyer	Chennai	15000.0	72000.0	South
3	104	Neha Singh	Delhi	15000.0	NaN	North
5	106	Karan Shah	Ahmedabad	18000.0	61000.0	West
6	107	Pooja Das	Kolkata	14000.0	NaN	East
7	108	Riya Kapoor	Jaipur	16000.0	69000.0	North

```
#3. Mention how many records were lost
# Answer= 1 record lost (Customer_ID 105).
```

Q9. Imputation

Handle missing values in Monthly_Sales using:

• Forward Fill

```
#1. Apply forward fill
df['Monthly_Sales']= df['Monthly_Sales'].ffill()
```

```
df
```

	Customer_ID	Name	City	Monthly_Sales	Income	Region
0	101	Rahul Mehta	Mumbai	12000.0	65000.0	West
1	102	Anjali Rao	Bengaluru	12000.0	NaN	South
2	103	Suresh Iyer	Chennai	15000.0	72000.0	South
3	104	Neha Singh	Delhi	15000.0	NaN	North
4	105	Amit Verma	Pune	18000.0	58000.0	None
5	106	Karan Shah	Ahmedabad	18000.0	61000.0	West
6	107	Pooja Das	Kolkata	14000.0	NaN	East
7	108	Riya Kapoor	Jaipur	16000.0	69000.0	North

```
#3. Explain why forward fill is suitable here
```

Q10. Flagging Missing Data

Create a flag column for missing Income

```
#1. Create Income_Missing_Flag (0 = present, 1 = missing)

df['Income_Missing_Flag'] = df['Income'].isna().astype(int)

print(df['Income_Missing_Flag'])
```

```
0    0
1    1
2    0
3    1
4    0
5    0
6    1
7    0
Name: Income_Missing_Flag, dtype: int64
```

```
#2. Show updated dataset
df
```

	Customer_ID	Name	City	Monthly_Sales	Income	Region	Income_Missing_Flag
0	101	Rahul Mehta	Mumbai	12000.0	65000.0	West	0
1	102	Anjali Rao	Bengaluru	NaN	NaN	South	1
2	103	Suresh Iyer	Chennai	15000.0	72000.0	South	0
3	104	Neha Singh	Delhi	NaN	NaN	North	1
4	105	Amit Verma	Pune	18000.0	58000.0	None	0
5	106	Karan Shah	Ahmedabad	NaN	61000.0	West	0
6	107	Pooja Das	Kolkata	14000.0	NaN	East	1
7	108	Riya Kapoor	Jaipur	16000.0	69000.0	North	0

```
#3. Count how many customers have missing income

Missing_Flag = df['Income_Missing_Flag'].value_counts()
Missing_Flag
```

Income_Missing_Flag	count
0	5
1	3

dtype: int64

```
Start coding or generate with AI.
```